INGRID VASCONCELLOS BUNHOLI

Ph.D. Candidate in Marine Science | The University of Texas at Austin Marine Science Institute

Port Aransas, Texas | ingrid.bunholi@utexas.edu | Google Scholar

PROFESSIONAL PROFILE

Marine molecular ecologist and conservation geneticist integrating population genomics, environmental DNA, community ecology, and quantitative approaches to study biodiversity across biological scales. Research examines species identification, population connectivity, ecological interactions, and community structure in reef fishes, estuarine fishes, sharks, and rays. Experienced in field and laboratory research, genomic data analysis, and interdisciplinary collaborations across institutions and countries.

EDUCATION

Ph.D., Marine Science **Expected November 2026**

The University of Texas at Austin Marine Science Institute, USA

Doctoral Discovery Fellowship (five years of support); GPA: 3.9

M.S., Biology (Ecology) **2022**

Indiana State University, USA

Teaching Assistant Scholarship, Department of Biology

M.S., Biology (Zoology) **2020**

Universidade Estadual Paulista (UNESP), Brazil

CAPES Research Scholarship

B.S., Environmental Engineering **2019**

Universidade Federal de São Paulo (UNIFESP), Brazil

CNPq Undergraduate Research Scholarship

B.S., Marine Sciences **2016**

Universidade Federal de São Paulo (UNIFESP), Brazil

FAPESP Undergraduate Research Scholarship

SELECTED APPLIED CONSERVATION CONTRIBUTIONS

- Developed an open-access rapid DNA/eDNA practitioner guide to improve monitoring of sharks and rays affected by fisheries and wildlife trade (English and Portuguese).
- Integrated environmental DNA with fisheries and resource-survey data to evaluate monitoring approaches and develop the first species-specific primer for Southern flounder (*Paralichthys lethostigma*) in partnership with Texas Parks and Wildlife Department (TPWD).
- Produced the first range-wide population-genetic study of the angular angelshark (*Squatina guggenheim*), providing evidence relevant to IUCN reassessment and conservation planning.
- Co-lead a Brazil-wide reef-fish DNA barcode gap analysis and reference-library initiative involving more than 15 researchers, field expeditions, and museum collections.
- Led interdisciplinary studies of cryptobenthic reef-fish ecology, host associations, microbiomes, and population connectivity.

RESEARCH EXPERIENCE

Doctoral Research Fellow **2022-present**

Casey Lab, The University of Texas at Austin Marine Science Institute

- Lead interdisciplinary research on the ecology, genomics, microbiomes, and monitoring of cryptobenthic reef fishes.
- Design and coordinate field sampling, molecular workflows, and bioinformatic pipelines across collaborative projects.
- Develop conservation-relevant approaches for marine species using eDNA/eRNA, population genomics, and trait-based ecological evidence.
- Mentor undergraduate researchers and coordinate collaborations across U.S. and international institutions.

Visiting Research Collaborator **2026**

Gatins Lab, Marine Evolutionary Genomics, Northeastern University

- Led molecular and bioinformatic workflows for the assembly and annotation of the first whole-genome resource for the genus *Eviota* using Oxford Nanopore technology.

Visiting Research Collaborator **2023**

Biodiversity Genomics Lab, Newcastle University

- Conducted population-genomic analyses of sponge-dwelling cryptobenthic reef fishes using ddRAD-seq.

Research Assistant **2015-2020**

Laboratório de Genética e Conservação, Universidade Federal de São Paulo

- Conducted molecular conservation research on endangered sharks and rays using DNA barcoding and population genetics.
- Contributed to studies of fisheries bycatch, illegal wildlife trade, and phylogeography in the Southwest Atlantic.

PUBLICATIONS

Ingrid V. Bunholi is shown in bold. *Equal contribution.

Manuscripts under review

Bunholi, I. V., Kordal, M., Souza, C. M., Olsen, Z., Mauvisseau, Q., & Casey, J. M. (2026). Integrating fisheries surveys and environmental DNA to improve monitoring of southern flounder (*Paralichthys lethostigma*) in Texas estuaries. *Estuaries and Coasts*. Under review.

Brandl, S. J., Foster, N. R., Semmler, R. F., **Bunholi, I. V.**, et al. (2026). Reframing the species-energy relationship: Energy quality, not quantity, promotes coastal marine biodiversity. *Nature Ecology & Evolution*. Under review.

Peer-reviewed articles

Bunholi, I. V., Affleck, A., Do, M. C., Johnson, K., Ortega, R., Jensen, E., Tornabene, L., Brandl, S. J., Casey, J. M., & Rueger, T. (2026). Ecological constraints of habitat specialization shape the social organization and life history of a sponge-dwelling goby. *The American Naturalist*.

Bunholi, I. V., Affleck, A., Johnson, K., Ortega, R., Freeman, C. J., Rueger, T., & Casey, J. M. (2026). Cryptic interactions in a marine microcosm: Sponges shape the microbiome of a resident goby. *Molecular Ecology* 35, no. 16: e70523.

Bunholi, I. V.*, Alvarenga, M.*, Adachi, A. M. C. L., Cruz, M. M., Feitosa, L. M., de Jesus, E. V., Lopes, M. E. L., Povil, C., Souza, D., Torres, Y., Solé-Cava, A. M., Domingues, R. R., Charvet,

P., & Cruz, V. P. (2025). Rapid DNA/eDNA-based ID tools for improved chondrichthyan monitoring and management. *Molecular Ecology Resources*, e70044.

Alvarenga, M., **Bunholi, I. V.**, Brito, G. R., Charvet, P., Domingues, R. R., Siqueira, M. V. B. M., & Cruz, V. P. (2024). Years of extensive and inadvertent consumption of endangered sharks and rays in Brazil unraveled through DNA analysis: How can we do better? *Biological Conservation*, 292, 110543.

Bunholi, I. V., Foster, N. R., & Casey, J. M. (2023). Environmental DNA and RNA in aquatic community ecology: Toward methodological standardization. *Environmental DNA*, 5, 1133-1147.

Chini, D. S., Mondal, N., Kar, A., **Bunholi, I. V.**, et al. (2023). Seasonal variability of marine fish diversity in relation to water quality of the East Midnapore Coast of West Bengal, India. *Ocean Science Journal*, 58(2), 13.

Bunholi, I. V., Ferrette, B. L. S., Domingues, R. R., Rotundo, M. M., Cuevas, J. M., García, M., et al. (2021). Multilocus phylogeography of the endemic and endangered angular angelshark (*Squatina guggenheim*) in the Southwest Atlantic Ocean. *Hydrobiologia*, 849(9), 2177-2192.

Banerjee, P., Stewart, K., Antognazza, C., **Bunholi, I. V.**, Deiner, K., Barnes, M., et al. (2021). Plant-animal interactions in the era of environmental DNA (eDNA): A review. *Environmental DNA*, 4(5), 987-999.

Domingues, R. R., **Bunholi, I. V.**, Pinhal, D., Antunes, A., & Mendonça, F. F. (2021). From molecule to conservation: DNA-based methods to overcome frontiers in the shark and ray fin trade. *Conservation Genetics Resources*, 1-17.

Ferrette, B. L. S., Domingues, R. R., Rotundo, M. M., Miranda, M. P., **Bunholi, I. V.**, Biasi, J. B., et al. (2019). DNA barcode reveals the bycatch of endangered batoid species in the Southwest Atlantic: Implications for sustainable fisheries management and conservation efforts. *Genes*, 10(4).

Bunholi, I. V., Ferrette, B. L. S., Biasi, J. B., Magalhães, C. O., Rotundo, M. M., Oliveira, C., et al. (2018). The fishing and illegal trade of the angelshark: DNA barcoding against misleading identifications. *Fisheries Research*, 206, 193-197.

Books and book chapters

Boza BR, **Bunholi IV**, Charvet P, Paes da Cruz V, Alvarenga MM (2025) From Sample to Rapid DNA and eDNA Identification: An Easy Step-by-Step Guide. UNESP *Botucatu, São Paulo,* ISBN 978-65-89398-39-4.

Bunholi, I. V. (2021). Cação-anjo - Tubarões ou realmente anjos? In Encontros e Desencontros com Tubarões e Raias: Uma História (Não) contada por Pesquisadores Brasileiros. *Associação MarBrasil*, ISBN 978-65-00-157642.

COMPETITIVE FUNDING, FELLOWSHIPS, AND AWARDS

Research funding and Awards

AAUS Foundation Doctoral Scholarship	$3,000 | 2026
Fisheries Society of the British Isles Small Research Grant	$5,500 | 2024
International Coral Reef Society Graduate Fellowship	$2,500 | 2024
eDNA Collaborative/Physalia Course Microgrant	$600 | 2024
Texas Chapter American Fisheries Society Clark Hubbs Student Scholarship	$1,500 | 2024
Centre for Biodiversity Genomics iBOL Conference Travel Award	$800 | 2024
College of Graduate and Professional Studies Graduate Student Research Grant *Indiana State University*	$800 | 2021
American Elasmobranch Society Award for Young Professional Fund	2021

Fellowships and scholarships

Doctoral Discovery Fellowship *The University of Texas at Austin Marine Science Institute; five years of support*	$190,000 | 2022
Department of Biology M.S. Scholarship *Indiana State University*	$36,000 | 2020
CAPES M.S. Scholarship *Universidade Estadual Paulista*	$7,000 | 2018
CNPq Undergraduate Research Scholarship *Universidade Federal de São Paulo*	$2,000 | 2017
FAPESP Undergraduate Research Scholarship *Universidade Federal de São Paulo*	$3,500 | 2016

Scientific peer review

Reviewer *for Molecular Ecology, Environmental DNA, Scientific Reports, PLOS ONE, Environmental Science and Pollution Research, Journal of Fish Biology, Neotropical Ichthyology, Reviews in Fish Biology and Fisheries, Oecologia Australis, Conservation Genetics Resources, Molecular Ecology Resources, and Communications Biology*

TEACHING AND OUTREACH EXPERIENCE

Teaching Assistant in Molecular Ecology, *UTMSI*	**2026**
Summer Science Camp Teacher (Grades 5-9), *UTMSI*	**2026**
Science in Residence Teacher (Grades 4-5), *UTMSI and Olsen Elementary School*	**2023**
Volunteer Teacher (Grades 5 and 7), *Skype a Scientist Program*	**2023**
Teaching Assistant in Biology, *Indiana State University*	**2021-2022**
Teaching Assistant in Molecular Biology, *Universidade Federal de São Paulo*	**2015-2016**

INVITED LECTURES AND CONFERENCE PRESENTATIONS

Invited speaker

2025 *Compreendendo o status e os desafios do DNA ambiental na conservação marinha* [Understanding the status and challenges of environmental DNA in marine conservation]. ElasmoTalks: 1º Encontro Virtual sobre Tubarões e Raias, UNESP, Brazil.

2025 *The power of genetic tools to understand biodiversity of coral reef fishes.* Texas A&M University College of Marine Sciences & Maritime Studies, USA.

2024 *O papel do DNA na Preservação da Biodiversidade* [The role of DNA in biodiversity conservation] Projeto TAMAR/ICMBio, Fernando de Noronha, Brazil.

Oral presentations

2025 *Gimme more than shelter: Sponge host shapes the skin and gut microbiome of a resident goby.* Annual Benthic Ecology Meeting, Mobile, Alabama.

2024 *From DNA barcoding to eDNA/eRNA metabarcoding: Understanding the status and challenges of molecular tools in marine conservation.* 9th International Barcode of Life Conference, Belém, Brazil.

2024 *Fine-scale genetic structure and diversification patterns of a sponge-dwelling goby*. Annual Benthic Ecology Meeting, Charleston, South Carolina.

2023 *Environmental DNA and RNA in community ecology across the tree of life: A methodological review*. 6th Annual Environmental DNA Technical Exchange Workshop.

2022 *Machine learning e conservação: Previsão de categorias de ameaça em elasmobrânquios com dados deficientes*. III ELASMulheres.

2021 *Genetic population structure of the angular angelshark, Squatina guggenheim, an endangered endemic species from the Southwest Atlantic Ocean*. Society for the Study of Evolution Virtual Meeting.

2019 *Genetic population structure of the angular angelshark, Squatina guggenheim, an endangered endemic species from the Southwest Atlantic Ocean*. Brazilian-International Congress of Genetics.

SCIENCE COMMUNICATION AND MEDIA

Popular articles

2024 Borbee, E., **Bunholi, I. V**., Changsut, I., & Good, A. Everything is Bigger in Texas: Introducing the Texas Chapter of the Society for Women in Marine Science. CERF's Up!, 50(3).

2024 **Bunholi, I. V.** Discovering the hidden allies beneath the waves: The sponge-fish connection. The Port Aransas South Jetty News - Institute Insights.

2023 **Bunholi, I. V.** Water analysis "goldmine" of information. The Port Aransas South Jetty News - Institute Insights.

Media coverage

2026 Alvarenga MM, **Bunholi IV**, Boza BR, Paes da Cruz V, Charvet P. Rapid DNA and eDNA tools are transforming shark and ray protection. *IUCN SSC Shark Specialist Group, Shark News*, Issue 13, 198–200.

2025 **Bunholi IV**. Fast solutions for an urgent problem: Genetic identification for the protection of sharks and rays. *The University of Texas Marine Science Institute*, September 19, 2025.

2024 Alvarenga M and **Bunholi IV** et al. Fifteen years of elasmobranch trade unveiled by DNA tools. Featured in *Forbes Science* by Melissa Cristina Márquez.

TECHNICAL AND FIELD EXPERTISE

Molecular ecology and genomics

DNA barcoding; environmental DNA and RNA; metabarcoding; species-specific qPCR and ddPCR assays; whole-genome sequencing (Oxford Nanopore); RADseq; population genetics and genomics; microbiome profiling; molecular assay design and validation.

Quantitative and computational

R, Python, and Bash; ecological statistics; multivariate community analyses; metabarcoding pipelines; population-genomic workflows; data visualization; reproducible analysis pipelines.

Software

QIIME2, STACKS, Geneious, and associated command-line bioinformatics tools for whole-genome assembly.

Field and diving

Scientific diving (+100 dives); reef fish visual surveys, marine fish and environmental-water sampling; reef and estuarine fieldwork; tissue collection and preservation.

CERTIFICATIONS

2023 American Academy of Underwater Sciences (AAUS) Scientific Diver

2023 CPR, First Aid, and Oxygen Administration - Divers Alert Network

2022 SSI Rescue and Stress Diver

2022 SSI Open Water Diver

PROFESSIONAL DEVELOPMENT

2026 Using droplet digital PCR (ddPCR) for environmental DNA and RNA applications, Natural *History Museum, Oslo, Norway* (40 hours)

2024 Environmental DNA metabarcoding, *Physalia Courses/eDNA Collaborative* (40 hours)

2023 RADseq data analysis using STACKS, *Physalia Courses* (40 hours)

2023 Marine invasive-species detection using molecular tools (eDNA), *OceanTeacher Global Academy* (80 hours)

LANGUAGES

Portuguese - Native | **English** - Advanced | **Spanish** - Basic